

Paideia Math Team Practice

Mass Points and Ceva's Theorem

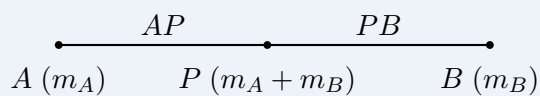
Mass Points

Mass point geometry assigns *masses* to points so that ratios of segment lengths can be handled like a balanced lever. If P lies on segment AB , assign masses m_A and m_B to A and B . Then

$$\frac{AP}{PB} = \frac{m_B}{m_A}.$$

So the segment lengths are inversely proportional to the masses at their endpoints. The mass at the balancing point is the sum of the endpoint masses:

$$m_P = m_A + m_B.$$



For a triangle with concurrent cevians, choose convenient masses at the vertices and carry the sums to points on the opposite sides. The same mass assigned to a point must be consistent along every line through that point.

In triangle ABC , points D , E , and F lie on BC , CA , and AB , respectively, and AD , BE , and CF are concurrent. Suppose AB , AC , and BC have lengths 13, 14, and 15, respectively. If

$$\frac{AF}{FB} = \frac{2}{5} \quad \text{and} \quad \frac{CE}{EA} = \frac{5}{8},$$

find BD and DC .

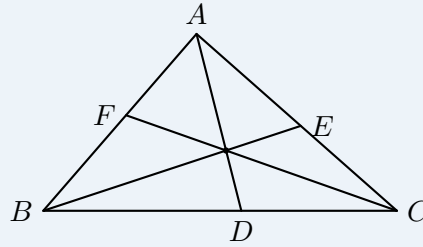
Triangle ABC has point D on AB , point E on BC , and point F on AC . Lines AE , CD , and BF intersect at point G . The ratio $AD : DB$ is $3 : 5$, and the ratio $CE : EB$ is $8 : 3$. Find the ratio $FG : GB$.

Ceva's Theorem

Let D , E , and F lie on sides BC , CA , and AB of triangle ABC . Then the cevians AD , BE , and CF are concurrent if and only if

$$\frac{BD}{DC} \cdot \frac{CE}{EA} \cdot \frac{AF}{FB} = 1.$$

For interior points, all of these ratios are positive. This theorem is especially useful when two of the three side ratios are known: solve for the third ratio directly. Mass points encode the same balance relationships and can also determine ratios along the cevians themselves.



In triangle ABC ,

$$AB = \frac{20}{11}AC.$$

The angle bisector of $\angle A$ intersects BC at point D , and point M is the midpoint of AD . Let P be the point of intersection of AC and line BM . The ratio $CP : PA$ can be expressed as $m : n$, where m and n are relatively prime positive integers. Find $m + n$.

Let P be an interior point of triangle ABC , and extend the lines from the vertices through P to the opposite sides. Let a , b , c , and d denote the lengths of the segments indicated in the figure. Find the product abc if

$$a + b + c = 43 \quad \text{and} \quad d = 3.$$

